a Pre-silicon model stack

workload models
Lorenz-96 ensemble, EnKF, 3D advection, microphysics, DA — Python, real numerics

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AVT cycle model
issue, SFU contention, SRAM banks, halo wait, precision switch → 71.1% sustained

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chip model
64 tiles, L2/L3/HBM contention, DVFS, leakage → 29.1 ms, 1.26 J per step

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baseline models
H100 / A100 rooflines with published efficiency ranges, power states

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thermal / cost / reliability
RC network, Monte-Carlo corners, Murphy yield, Weibull life

b Experiment matrix

id	experiment	output
E1	kernel throughput	time, utilisation
E2	energy breakdown	J per block
E3	ablation	feature attribution
E4	Monte-Carlo	ratio distribution
E5	precision sweep	skill vs energy
E6	thermal transient	junction trace
E7	corner analysis	speed yield
E8	yield & cost	$ per die

c What the model is calibrated against

quantity	source	match
fp32 FMA energy	Horowitz 2014, scaled	0.42 pJ
SRAM / HBM pJ/byte	published 3 nm / HBM3E	±25%
stencil efficiency	WRF / ICON GPU studies	22–50%
idle & TDP	H100 SXM5 data sheet	90 / 700 W
Lorenz-96 skill	reference EnKF	0.22 RMSE

Every assumption is a named constant in `spec/acpu1.json`; the Monte-Carlo study in Fig. 14 reports the resulting 90% interval rather than a single point.